Yusef Syed

Software Engineer: AI Agents, Security & Developer Tools

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Toronto, ON | U.S. & Canadian citizen, no sponsorship needed | Will relocate to San Francisco or New York

Available full-time from January 2027 (taking leave from U of T; open to making it permanent)

Education

University of Toronto **Expected May 2030**

B.Sc., Mathematical & Physical Sciences (first year); intended Math and CS double major *Toronto, ON*

Coursework: Linear Algebra, Proofs & Calculus, Statistics; Data Structures (LCCC, dual enrollment)

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, Swift

AI & security: LLM agents, LangGraph, Inspect, Promptfoo, PyRIT, PyTorch; access-control testing (IDOR, RLS), OWASP LLM/Agentic Top 10, MITRE ATLAS, CWE

Systems & tools: Node.js, React Native, SwiftUI, PostgreSQL, Supabase, Docker, GitHub Actions, Git, Claude Code

Experience

Scopehaven | scopehaven.dev | github.com/YusefSyed/scopehaven-oss **Oct 2026 – Present**

Founder & Engineer (solo project); security testing for AI agents *Toronto, ON*

- Built **Scopehaven**, a Node.js CLI and GitHub Action that tests whether an AI agent touched the wrong account, tenant or refund, decided from the app's own records rather than the agent's report.
- Released the **Apache-2.0 open edition** (v0.7.0; Node 22 and 24, 4,800+ tests, zero dependencies): MCP test target, GIF demo; on our synthetic test app the vulnerable build fails 26 of 44 checks.
- Designed its evidence format (DSSE/in-toto exports, an RFC 9162 hash-chained ledger, a countersignature with an RFC 3161 timestamp) and a stdlib-Python checker that re-derives verdicts from the kept records: **88 of 88** re-derived on the sample pack (synthetic app, no model), 8 of 8 tampered copies rejected.
- Built **16 incident-class checks** from a corpus of public agent-failure reports; on our own test lab each fails 3 of 3 on a build with that failure planted and passes on the clean build.
- Ran the merge gate for parallel AI coding agents as the only human on the repo: exact-head CI, a recorded review by a second session and secret scanning; 57 review verdicts asked for changes, 0 git reverts, 1 regression restored in 20 h.

Independent Security Research **Oct 2026 – Present**

Security Researcher *Remote*

- Filed **8 private security advisories** (authorization, credential and data-exposure flaws; one rated High in my report) with 5 open-source projects, each reproduced 3/3 with controls first; all in maintainer triage.
- Ran an internal pilot on an open-source AI workspace (synthetic data): after a user was suspended, an in-flight agent run still wrote a file in **2 of 14** cases; a partial local patch passed 14/14 on the tested path.

Open Source

14 merged PRs: Microsoft, Apache Arrow, W&B, Inspect Scout, Kornia | Full list **2026**

- Microsoft PyRIT (AI red-teaming): added package-hallucination attack techniques for Dart, Perl and Raku.
- W&B RAI Toolkit: landed five evaluation fixes, including a concurrency fix the maintainer verified at 3.34s to 1.10s.
- Microsoft Agent Lightning: landed four reliability fixes, e.g. case-insensitive auth-header replacement.

Projects

effect-tracer | *Rust, Linux ptrace/seccomp, Ed25519* | signed v0.1.2 release with SBOM and VEX | Repo **2026**

- Built a tracer that signs a process tree's file writes, connects, DNS and execs into a hash chain: **19 of 20** seeded bypasses caught at 1.1–1.9x overhead with seccomp (20 of 20 at 5–298x without); 0 undeclared events in 30 clean runs.

ToolEffectBench | *Python, Inspect, Docker, SQLite* | Repo **2026**

- Built an Inspect benchmark scoring prompt injection from persisted database state, not transcripts (188 samples, 17 models; missing evidence never safe). Found and disclosed a defect hiding attack text from 10 models; reran those arms.

Agent Eval Mutation Lab | *Python, Inspect* | **21 of 21** decided verdicts matched an independent SQLite oracle **2026**

Providence | *Swift* | **1st Place, Aramco Americas Challenge** (Best Beginner Hack), Hack the North 2026 **2026**

- Built the Apple Watch client for a team of four's computer-control AI agent: spoken requests, approvals, verified results.

Aesthetics AI & **Tiraz** | *TypeScript, React Native* | two iOS apps on the **App Store** | Study **2026**

Awards & Activities

AP Scholar with Distinction | **USA Powerlifting:** #2 in Pennsylvania, #19 in the U.S. (60 kg, 2025) | **Feeding the Valley**, Vice President (2025–26): 60+ volunteers, $20,000 raised, 1,700 meals for food banks